

HOMOLOGY AND COHOMOLOGY: Whenever one associates to a topological space X a chain complex

$$\cdots \rightarrow C_{n+1} \xrightarrow{\partial_{n+1}} C_n \xrightarrow{\partial_n} C_{n-1} \rightarrow \cdots,$$

we obtain a well-defined **homology theory** (simplicial, singular, cellular, etc) by defining the n -**th (adjective) homology group** as $H_n(X) = \ker \partial_n / \operatorname{im} \partial_{n+1}$. Fixing some abelian group G (for us for now, $G = \mathbb{Z}$) gives a well-defined cohomology theory via the co-complex

$$\cdots \rightarrow \operatorname{Hom}(C_{n-1}, G) \xrightarrow{\partial_n} \operatorname{Hom}(C_n, G) \xrightarrow{\partial_{n+1}} \operatorname{Hom}(C_{n+1}, G) \rightarrow \cdots,$$

where $\partial^n : f \mapsto f \circ \partial_n$. The n -**th (adjective) cohomology group** is $H^n(X) = \ker \partial^{n+1} / \operatorname{im} \partial^n$.

(CO)HOMOLOGY FACTS:

- (1) H_n and H^n are functors. In particular, a continuous map $X \rightarrow Y$ yields maps $H_n(X) \rightarrow H_n(Y)$ and $H^n(Y) \rightarrow H^n(X)$.
- (2) If the maps f and g are homotopy equivalent, the induced maps on (co)homology are equal.
- (3) If two different (co)homology theories can be applied, they will give the same answer. E.g., writing the sphere as a simplicial or as a CW complex doesn't change (co)homology.

DEFINITION/THEOREM: To get **reduced homology** $\tilde{H}_*$, augment your existing complex by setting $C_{-1} = \mathbb{Z}$ and mapping each generator of C_0 to $1 \in \mathbb{Z}$. It is a fact that $H_0(X) = \tilde{H}_0(X) \oplus \mathbb{Z}$, and $H_n(X) = \tilde{H}_n(X)$ for $n > 0$. Get **reduced cohomology** by applying $\operatorname{Hom}(-, \mathbb{Z})$ to the complex first.

DEFINITION:

- (1) A **0-cell** is a point in Euclidean space. For $n \in \mathbb{N}$, an n -**cell** is a subspace of $\mathbb{R}^n$ homeomorphic to the closed unit disk $\overline{D}^n = \{x \in \mathbb{R}^n \mid \|x\| \leq 1\}$. Note that this makes the boundary of an n -cell $\partial \overline{D}^n$ homeomorphic to $S^{n-1} = \{x \in \mathbb{R}^n \mid \|x\| = 1\}$.
- (2) A **CW complex** $X = \bigcup_n X_n$ is built as the union of n -skeletons denoted X_n for $n \in \mathbb{N}$ where:
 - (a) The **zero skeleton**, X_0 , is a discrete collection of points $\{e_\lambda^0\}_{\lambda \in \Lambda_0}$ that we call 0-cells.
 - (b) For $n \geq 1$ the n -**skeleton** X_n is built from X_{n-1} inductively by attaching a collection of n -cells $\{e_\lambda^n\}_{\lambda \in \Lambda_n}$ to the X_{n-1} skeleton via attaching maps $\varphi_\lambda : \partial e_\lambda^n = S^{n-1} \rightarrow X_{n-1}$. More precisely, one can consider X_n as the quotient

$$X_n = \left(X_{n-1} \bigsqcup_{\lambda \in \Lambda_n} e_\lambda^n \right) / \sim$$

where $x \sim \varphi_\lambda(x)$ for $x \in \partial e_\lambda^n$ for all $\lambda \in \Lambda_n$ [and endow X_n the quotient topology].

DEFINITION: Any (continuous) map $f : S^n \rightarrow S^n$ induces a group homomorphism $f^* : H_n(S^n) \rightarrow H_n(S^n)$, i.e., $f^* : \mathbb{Z} \rightarrow \mathbb{Z}$, uniquely determined by $1 \mapsto d$. This integer d is the **degree** of f .

DEFINITION: Given a CW complex X , define the **cellular chain complex**

$\cdots \rightarrow C_n \rightarrow C_{n-1} \rightarrow \cdots$ via C_n is the free abelian group generated by the n -cells of X .

The map $\partial : C_n \rightarrow C_{n-1}$ is defined via n -cell $[e_\lambda]$ maps to $\sum_{\mu \in \Lambda_{n-1}} d_{\lambda\mu} [e_\mu]$, where $d_{\lambda\mu}$ is the degree¹ of the attaching map. More precisely, $d_{\lambda\mu}$ is the degree of the map $\varphi_{\lambda\mu}$ defined via

$$\varphi_{\lambda\mu} : S^{n-1} \cong \partial e_\lambda \xrightarrow{\varphi_\lambda} X_{n-1} \twoheadrightarrow X_{n-1} / (X_{n-1} \setminus e_\mu) \cong S^{n-1}$$

The **cellular homology** of X is the homology coming from this complex.

¹Not circular! Use simplicial or singular homology to compute the degree of the attaching map. (Or, differential topology??)

- (1) Verify that for the co-complex for cohomology, the differential squares to zero.
- (2) Some useful algebra reminders when computing (co)homology.
- (a) If $f : V \rightarrow W$ is a $\mathbb{Z}$ -module homomorphism (equivalently, a homomorphism of abelian groups) between free (finitely generated) $\mathbb{Z}$ -modules V and W (that is, $V \cong \mathbb{Z}^n$ and $W \cong \mathbb{Z}^m$ for some $n, m \in \mathbb{N}$), show that $\text{rank}(V) = \text{rank}(\ker f) + \text{rank}(\text{im } f)$.^{2 3}
 - (b) If M is the matrix representation of a map $f : \mathbb{Z}^n \rightarrow \mathbb{Z}^m$, convince yourself that M^T (the transpose) is the matrix representation of the induced map $f^* : \text{Hom}(\mathbb{Z}^m, \mathbb{Z}) \rightarrow \text{Hom}(\mathbb{Z}^n, \mathbb{Z})$ where $f^*(\alpha) = \alpha \circ f$.
- (3) In this problem, we'll geometrically describe the coboundaries and cocycles of a simplicial complex.
- (a) Explain why there are no 0-cocycles.
 - (b) The 0-coboundaries are things (= "functions on vertices") in the kernel of the map from $C^0 \rightarrow C^1$, so we can focus on the underlying graph. Describe the 0-coboundaries when the graph is... (i) a single edge, (ii) a triangle, (iii) two disconnected edges. What is the general pattern?
 - (c) The 1-cocycles are things (= "functions on edges") in the image of the map from $C^0 \rightarrow C^1$, so we can focus on the underlying graph. Describe the 1-cocycles when the graph is... (i) a single edge, (ii) a path of length n , (iii) a triangle, (iv) a cycle of length n . What is the general pattern?
 - (d) The 1-coboundaries are things (= "functions on edges") in the kernel of the map from $C^1 \rightarrow C^2$. Describe the coboundaries when the simplicial complex is... (i) an empty triangle, (ii) a filled-in triangle. What is the general pattern?
 - (e) (Harder!) Can you figure out how to describe the 2-coboundaries and 2-cocycles?
- (4) Verify using simplicial (co)homology that
- $$\tilde{H}_k(S^n) \cong \tilde{H}^k(S^n) \cong \begin{cases} \mathbb{Z} & k = n \\ 0 & \text{else.} \end{cases}$$
- (5) Compute the cellular homology and cellular cohomology of a line segment. That is, consider the CW complex consisting of two 0-cells glued to a 1-cell at each end.
- (6) Compute the cellular homology and cellular cohomology of a 2-disk. Use the CW Complex structure which glues both ends of a 1-cell to a 0-cell and glues a 2-cell to fill the interior of the S^1 we just created.
- (7) Compute the cellular homology and cellular cohomology of a 2-sphere. Consider using the CW decomposition of gluing a 2-cell along the boundary to a 0-cell.
- (8) Compute the cellular homology & cohomology of $\mathbb{RP}^2$, using the CW structure where $X_0 =$ a point; $X_1 =$ attach one 1-cell, both endpoints attach to the point; $X_2 =$ attach one 2-cell, wrapping two-to-one around X_1 .

²Hint 1: Recall that a submodule of a free module over a Noetherian ring is necessarily free, and $\mathbb{Z}$ is Noetherian.

³Hint 2: Does the short exact sequence $0 \rightarrow \ker f \hookrightarrow V \twoheadrightarrow \text{im } f \rightarrow 0$ split?